

LLM 2-oy Goal

Goal: Bu oyda boshqa modellarni **download** qilib va **fine tuning** qilmaymiz. Balki **End to end LLM model** yasaymiz.

Uzimizni LLM-Model imizi hosil qilish **ketma-ketligi** (setup, 1~9):

🏆 Setup

Tayyorgarlik bosqich. Setup uchun **Google Colab** (GPU) kerak buladi

1. Data

Data ni qayerdan topamiz? Qanaqa sifat bulishi kerak? Qanchaga tushadi?

2. Data Preprocessing

Clean → Filter → Deduplicate

SML, DL, CV dagi preprocessing bilan deyarli bir hil, lekin ozgina farq qiladi

3. Tokenizer

Train BPE, Vocabulary

4. Dataset

Token IDs, Sequences, Batches

5. Model

Embedding, Attention, FFN, Transformer, (in pre-norm case)

6. Pretraining (training jarayon)

Next-token Prediction, Loss, Backpropagation Optimization

7. Evaluation

Loss Function, Perplexity, Generation, Benchmark

Benchmark - shunga uxshagan datalar bilan train qilingan boshqa model bilan qiyoslash

8. Inference

Load checkpoint, Generate next token, decode before output.

9. Deployment

API, Server, Docker, Local/Cloud

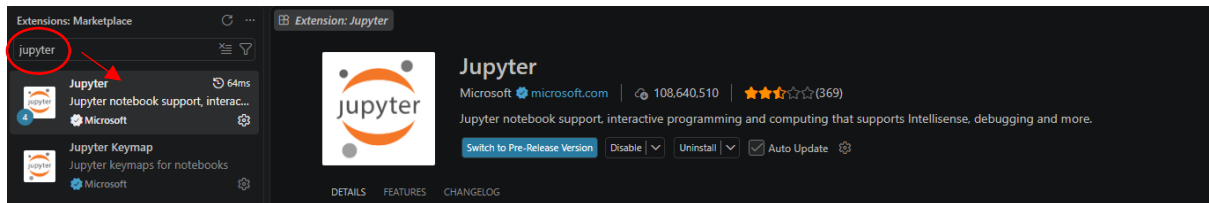
Deployni faqat AWS bilan emas, boshqa yangi va osonroq holat bilan ham kurib

Chiqamiz

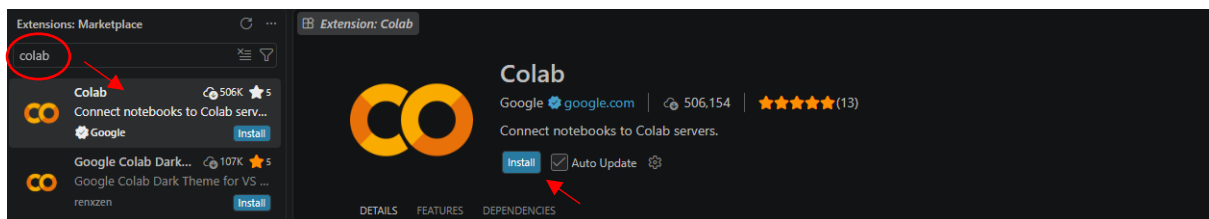
Connect - Google Colab with VS Code

Google Colab ni ishlatishdan maqsad – bizga training/pretraining uchun GPU kerak buladi

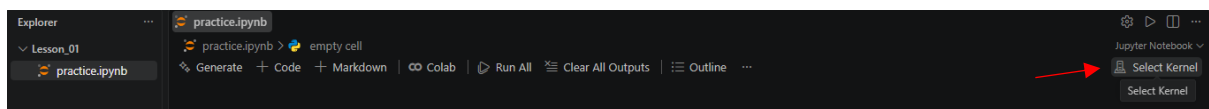
1. Jupyter Extensionini install qilamiz



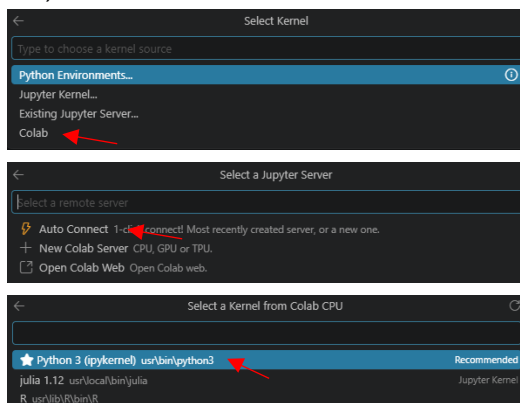
2. Google Colab Extensionini install qilamiz



3. Bitta .ipynb file ochamiz, va tepada "Select Kernel" buladi



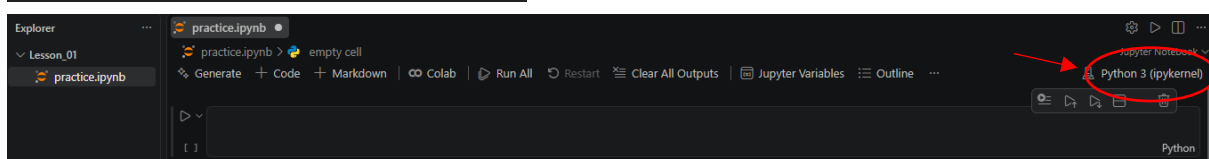
3.1) Select Kernel Click



Colab click

Auto Connect click (Colab hozir CPU da turipti)
Automatic ravishda Google Colabga utadi, va
email bilan tastiqlash kerak buladi

Colab click



```

import platform
import psutil
import torch

# 1. Print the operating system hosting your code
print(f"🖥️ Cloud OS Platform: {platform.platform()}")

# 2. Check the massive amount of free RAM Google assigned to you
total_ram_gb = psutil.virtual_memory().total / (1024 ** 3)
print(f"💾 Assigned Cloud RAM: {total_ram_gb:.2f} GB")

# 3. Check if your system detects the available GPU resources
gpu_available = torch.cuda.is_available()
print(f"🔥 Cloud GPU Available: {gpu_available} ({torch.cuda.get_device_name(0) if gpu_available else 'None'})")

```

Agar Colab ni connect qilishdan oldin shu codeni yozsak:

```

🖥️ Cloud OS Platform: Windows-10-10.0.26200-SP0
💾 Assigned Cloud RAM: 31.77 GB
🔥 Cloud GPU Available: True (NVIDIA GeForce RTX 3060)

```

Platform – Windows ekanini, RAM – 31.77 GB ekanini, va GPU bor!!

Agar Colab ni connect qilib keyin bu codeni yozsak:

```

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```

Shu turishda run qilsak

```

🖥️ Cloud OS Platform: Linux-6.6.122+-x86_64-with-glibc2.35
💾 Assigned Cloud RAM: 12.67 GB
🔥 Cloud GPU Available: True (Tesla T4)

```

Platforma – Linux ekanini, RAM – 12.67 GB ekanini, va GPU ulangan ekanini kursatadi

GPU vs TPU Comparison

Xususiyat	GPU (NVIDIA)	TPU (Google Custom Silicon)
Code Syntax	PyTorch, CUDA	PyTorch/XLA, JAX, TensorFlow
Qaysi holatda?	Learning, debugging, prototyping, va custom model yasayatkanda	Modellarni juda katta parameterlar bilan uqitkanda

Ecosystem	AWS, Colab, Local, Azure	Faqat Google Colab
Debugging	Ketma-ketlik oson	Murakkab
Real Use Case	• Universitet/startup 1B parameterli custom model • Local fine-tuning • Yangi model/prototip yaratish	• Juda katta LLM larni training qilish • Anthropic scaling • E-commerce recommendation tizimlari

Hardware Resources & Limits

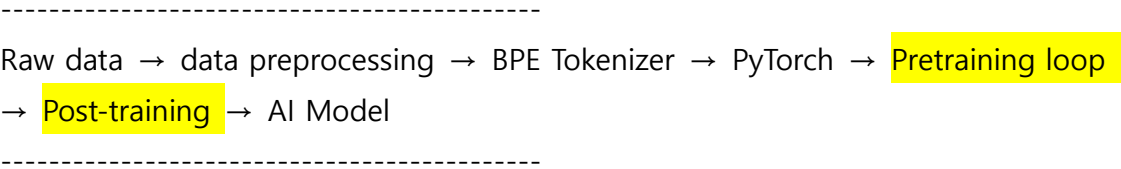
Hardware type	Idle Timeout	Max Session	System RAM	Credit
Standard CPU	90 minute	12 soat	~12.7 GB Basic	Tekin
T4 GPU	30 minute	12 soat	~15.0 GB Upgraded	Bergan soatga qarab
Cloud TPU	15 minute	12 soat	~47.0 GB Premium	Katta hisob

Idle Timeout: berilgan minute gacha ishlata ololadi. Va session utib ketsa tugaydi

Pretraining & Post-training

Pretraining: next token predict, output

Post-training: SFT, RLHF, DPO + RAG. After pretraining



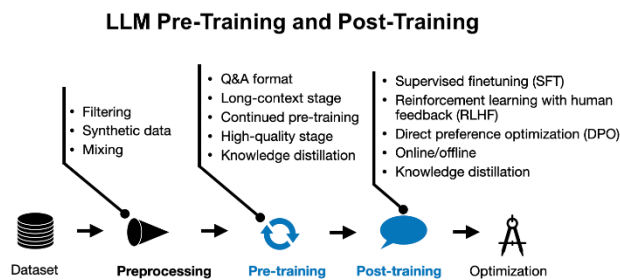
Use case: modelni yanadaham rivojlantirib, aqilli qilish uchun.

[**Pretraining**]:

Data Engineering



birinchi pre-training buladi va undan
keyin post-training buladi.



Tekinga student yokida budgeti kam odamlar uchun LLM model setup:

